

# **Confirmation of the relationship between stock market parameters and interbank credit market on the example of the Kazakhstan Stock Exchange.**

Magomet Yandiev <sup>1</sup>

Lomonosov Moscow State University, Faculty of Economics

Altana Andzhaeva <sup>2</sup>

Lomonosov Moscow State University, Faculty of Economics

*This article presents the calculations confirming practical applicability of earlier formulated theoretical model explaining relationship between the rate of one-day credits in the interbank market, volume of speculative investments and total securities under which transactions have been closed. This article is written based on the Kazakhstan stock exchange data<sup>3</sup>.*

## **1. Review of literature**

This article represents continuation of a series of the studies devoted to an examination of the relationship between the rate of one-day credits in the interbank market and some parameters of stock market, as well as to the proof of this relationship.

In the original article (Yandiev, 2011) the following formula is derived:

— —

- $u$  is the mean loss per a deal involving one stock;
- $I$  is the volume of speculative investments (amount of money on accounts in the authorized bank to the stock exchange and intended for speculations);
- $R$  is the rate of one-day credits in the interbank market, in fractions;
- $U$  is the total amount of stocks involved in deals;

The logic of the formula is as follows: the rate of one-day credits in the interbank market is inversely proportional to the number of securities traded on the stock exchange. The formula is purely theoretical as for its proof some assumptions and assumptions were used, but because of its simplicity, it is quite suitable for implementation of practical calculations. According to the logic of the formula, it can be considered workable in practice if the parameter  $u$  remains constant during calculations.

---

<sup>1</sup> Magomet Yandiev, mag2097@mail.ru

<sup>2</sup> Altana Andzhaeva, altana.andzhaeva@mail.ru

<sup>3</sup> Authors express their gratitude to the management of the Kazakhstan stock exchange and personally to Ms. Zarina Konkasheva for assistance in receiving necessary input data.

Calculations in paper (Pakhalov, Yandiev, 2013) were carried out on the basis of data received from the Moscow Stock Exchange. In another paper (Matveev, 2014) calculations were carried out based on the Bahrain Stock Exchange. In both cases, positive results were obtained, indicating that the formula correctly reflects the relationship of the parameters for the studied time intervals.

It should be noted that exchanges usually prefer not to disclose some parameters of the formula – the amount of clients' money and number of securities deposited within the exchange system. This situation is clear – disclosure of this information under certain circumstances may be a bad marketing capable to undermine investor confidence to validity of the quotations received at the exchange. On the other hand, general lack of such information in a free access only aggravates consequences of quite common situations when the quotation of particular issuer is formed at the exchange during the course of trading of absolutely scanty number of actions.

In the present article, we test the formula using the data of Kazakhstan stock exchange for the period 2010 – 2014.

## **2. Input data**

In order to test the applicability of the formula, the following data provided by Kazakhstan Stock Exchange were used (on a daily basis, for the period 2010-2014.):

- total amount of money deposited within the exchange system in m. tenge (analogue of I parameter, refer to Appendix 1);
- number of stocks (blue chips) deposited in the clearing exchange system, in pcs (U parameter, refer to Appendix 2);

We examined data on 10 most liquid stocks traded on KASE rather than on all of them, i.e. blue chips: Bank CenterCredit, Kazkommertsbank, KEGOC, Kazakhtelecom, KazMunaiGas Exploration Production, KazTransOil, KAZ Minerals, Kcell, Halyk Bank, Eurasian Natural Resources Corporation;

- fraction of blue chips in the total volume of stock trading, % (this information is needed to be sure that blue chips data is representative and reflect the situation on the stock market, refer to Appendix 3);
- rate of one-day credits in the interbank market, % a year (R parameter, refer to Appendix 4);
- number of securities involved in the stock exchange deals (as the analogue and substitute for the “number of all deposited stocks within the exchange system”, refer to Appendix 5).

Verification of practical applicability of the formula is performed as follows. The parameter  $u$  is calculated for every day during the entire analyzed period (1232 trading days for 2010-2014). Next, we use two different approaches. The formula will be considered correct if the parameter  $u$  has the minimum volatility (the first approach). The formula will be considered correct if the constructed regression equation corresponds to the theoretical model (the second approach)

At the same time in both approaches, the parameter  $U$  is substituted in two ways. As the quantity of all deposited stocks within the exchange system (the first option, the main) and as the quantity of securities involved in the stock exchange deals (the second option).

It is noteworthy that substantially more securities are deposited in the exchange system, than it is necessary for daily trading, 5000 times approximately (refer to Appendix 6). This reserve provides the Kazakhstan Stock Exchange an extremely high degree of resistance to possible surge in demand for the shares.

### **3. First approach. Formula verification based on standard deviation of the “ $u$ ” parameter**

The purpose of the first approach is to make sure that the standard deviation of parameter  $u$  is insignificant. We calculated mean and standard deviation for both options and plotted graphs for visual analysis of parameter  $u$  dispersion degree.

On the basis of performed calculations one can draw the following conclusions:

- If we compare the standard deviation of parameter  $u$  with the average value of the parameter  $u$  for the entire period of our analysis, it should be noted that the range of values of the parameter  $u$  looks rather wide, but if we compare the standard deviation with the average quotation per share, the volatility of the parameter  $u$  seems insignificant value (refer to Appendix 7).
- From a visual assessment of the  $u$  parameter dispersion, it is obvious that in general it is insignificant (refer to Appendices 8, 9 – parameter  $u$  is shown in historical sequence and to Appendices 10, 11 – parameter  $u$  is shown after sorting "from bigger to smaller").

Thus, it can be argued that the parameter  $u$  has low volatility and can be considered as a value close to a constant.

### **4. Second approach. Formula verification based on linear regression**

This approach involves the use of regression analysis of time series in order to identify relationships between the model parameters and check them for compliance with the theoretical model under consideration.

Input set of data consists of 1232 observations for each of six variables (refer to Appendix 12). Calculations were performed in the Gretl econometric package.

Since the regression analysis of time series requires all variables to be stationary, the first stage of econometric analysis involves an augmented Dickey-Fuller test (ADF) for each of the variables. Lag length in each case was set based on the Schwarz information criterion (SIC). All time series were examined for stationarity excluding trend. Results of tests are given in Appendix 13.

ADF test has shown that all variables except  $u\_big\_dep$  are stationary, therefore variable has to be tested for cointegration. According to Verbeek M., the existence of cointegration between the variables allows to get super consistent estimates of the model parameters, and the received results will make sense. Residuals of both regressions based on the deposited quantity and trading volume are stationary at the 1% level of significance (refer to Appendix 14). It allows us to formulate some conclusions from these regressions:

- The first regression equation is in general significant, as well as all its variables. The second equation is in general insignificant, and only one variable in it is significant at the 10% level of significance, which implies that the option of U calculation as the number of securities involved in the stock exchange deals is unreliable, and the impact of the variables included in the equation on the dependent variable may not even exist.
- Despite this, in both regression equations the I and R variables have positive coefficients, and the variable U has negative coefficient that completely corresponds to the logic of theoretical model.

Thus, the regression analysis confirms the significance of the tested formula.

## 5. Summary

Results of calculations for both options prove that the tested formula as a whole accurately reflects the relationship of parameters of the interbank credit market and the stock market.

Calculation of parameter U as the number of all deposited stocks within the exchange system is more correct, than understanding under it the number of securities involved in the stock exchange deals.

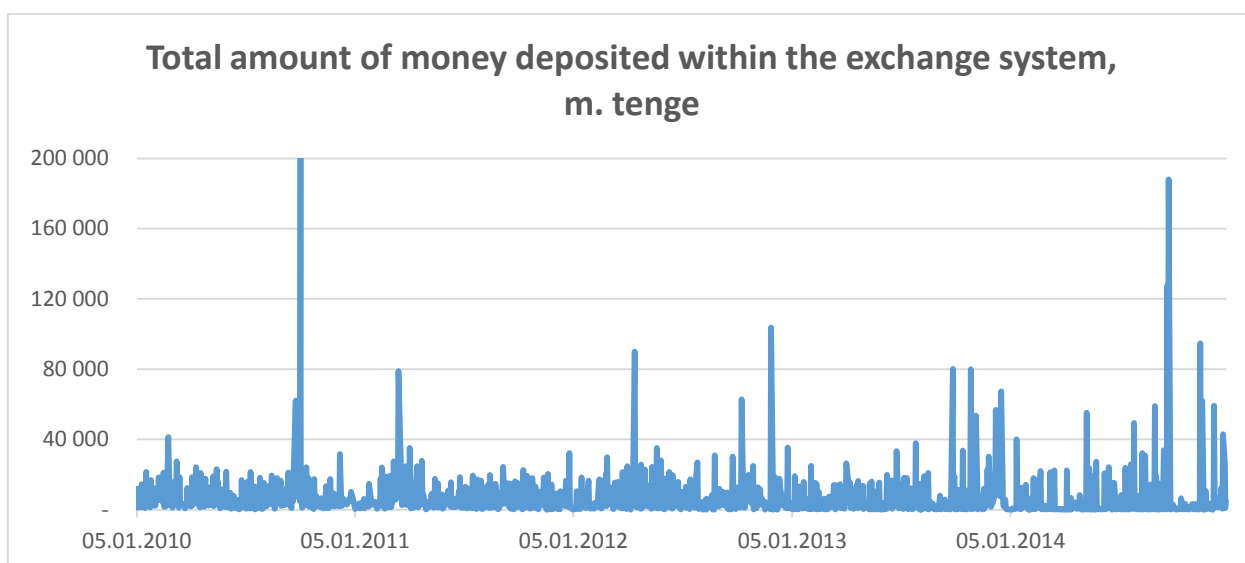
The findings of this work are consistent with the conclusions obtained in previous similar studies in Moscow (Pakhalov, Yandiev, 2013) and Bahrain stock exchanges (Matveev, 2014).

## 6. References

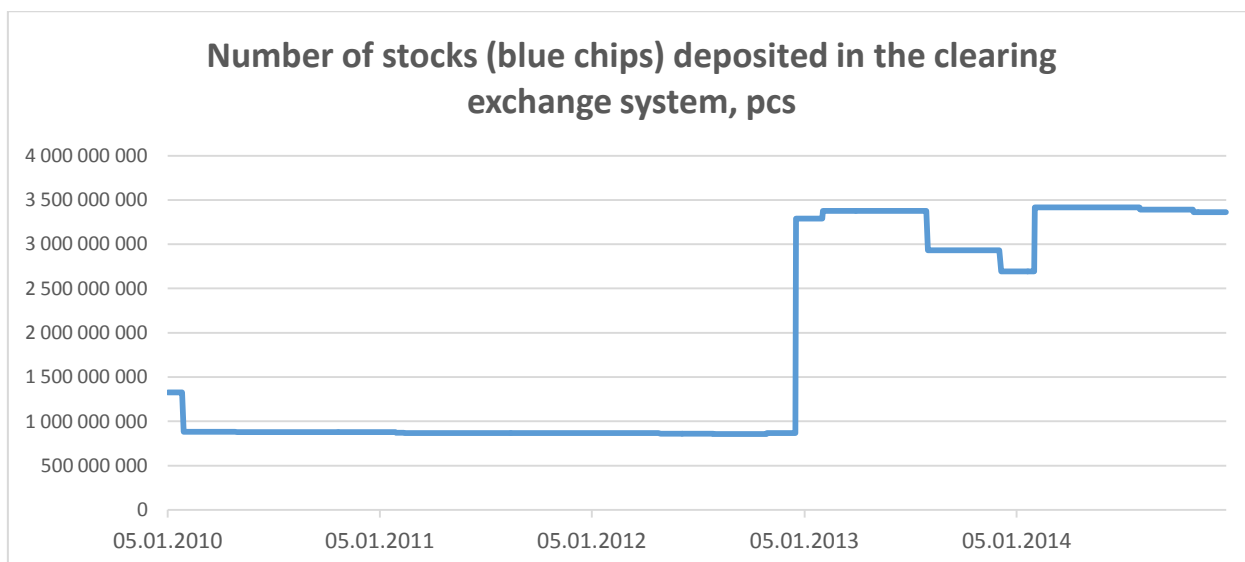
- Yandiev M. The Damped Fluctuations as a Base of Market Quotations // Economics and Management. 2011. N 16. URL: <http://ssrn.com/abstract=1919652>
- Yandiev, Magomet and Pakhalov, Alexander, The Relationship between Stock Market Parameters and Interbank Lending Market: An Empirical Evidence (September 23, 2013). Available at SSRN: <http://ssrn.com/abstract=2329871>
- Matveev, Aleksandr (2014) "Proving The Association Between Stock Market And Interbank Lending Market Parameters: The Bahrain Stock Exchange". Journal of Russian Review (ISSN 2313-1578), VOL. (0), 21-32. Available at: <http://rusreview.com/journal/vol-0-2014/14-proving-the-association-between-stock-market-and-interbank-lending-market-parameters-the-bahrain-stock-exchange-aleksandr-matveev.html>
- Verbeek M. A Guide to Modern Econometrics. 2nd ed. Chichester, 2004.

## Appendices 1-6. Input data

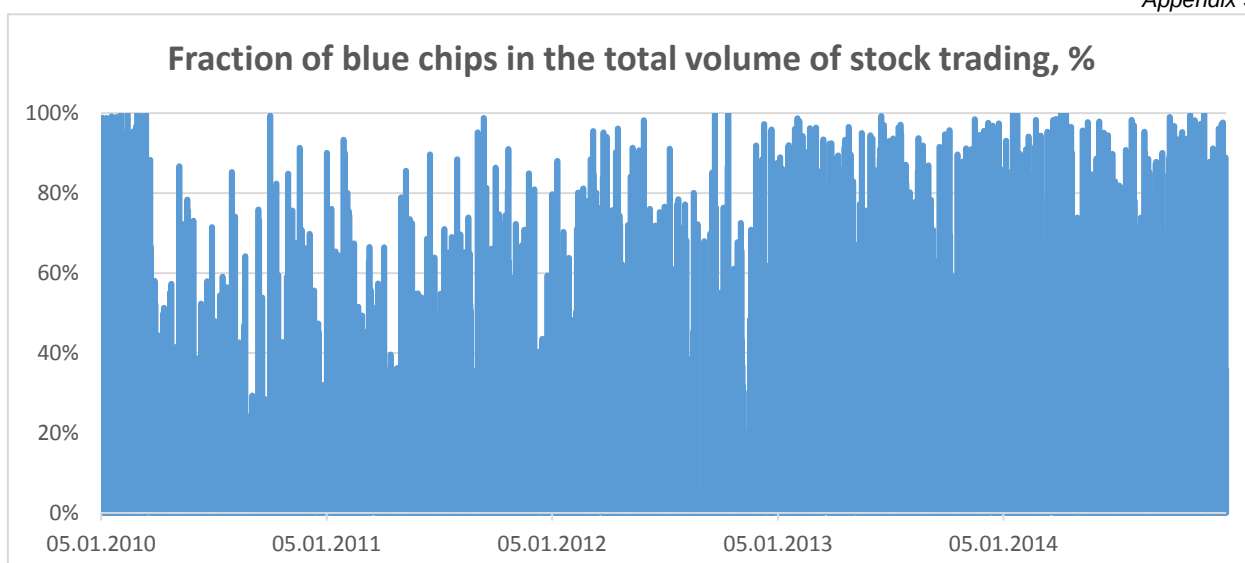
Appendix 1

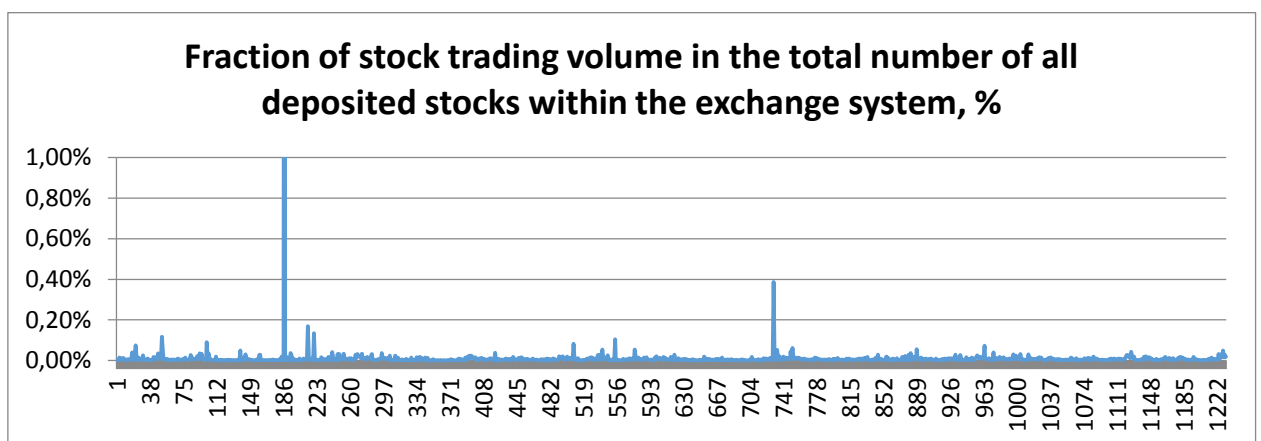
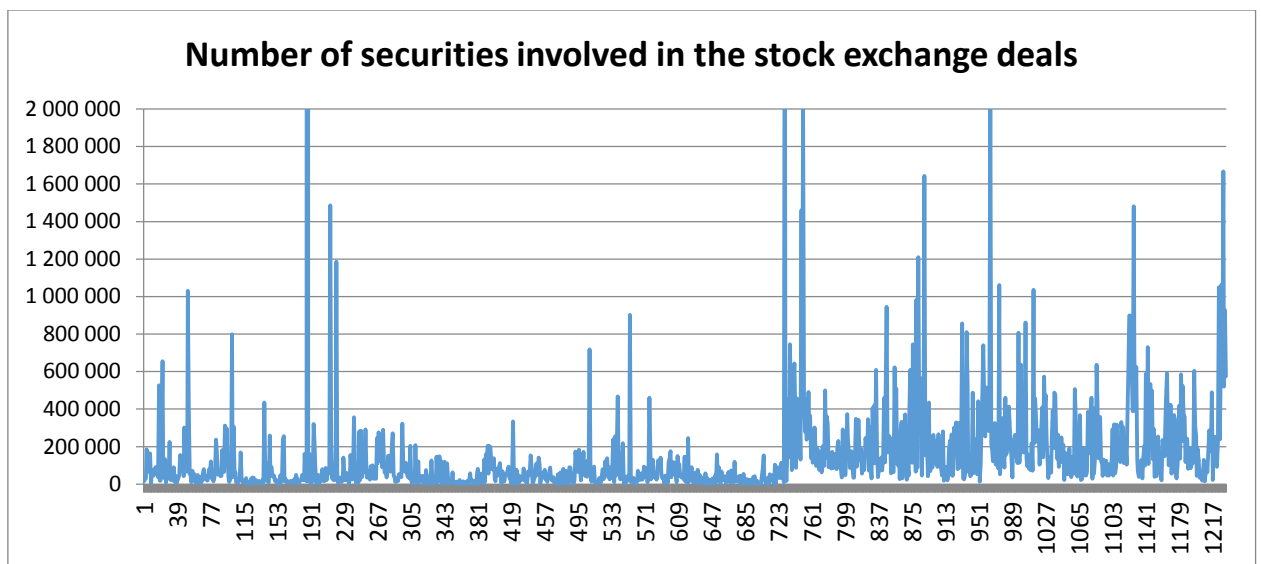
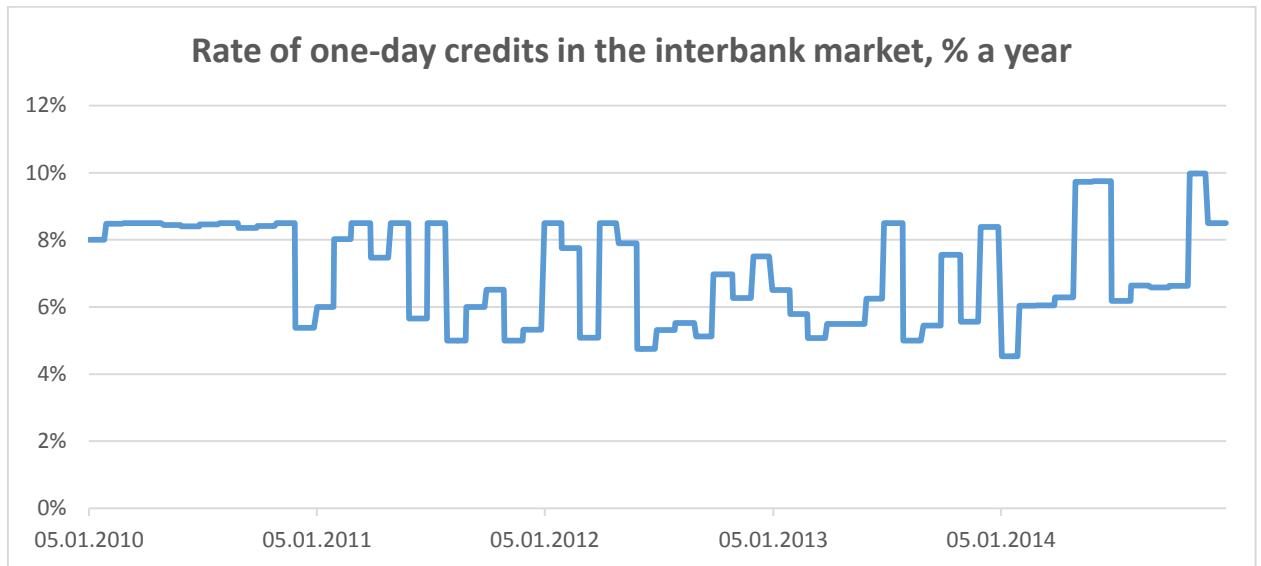


Appendix 2



Appendix 3





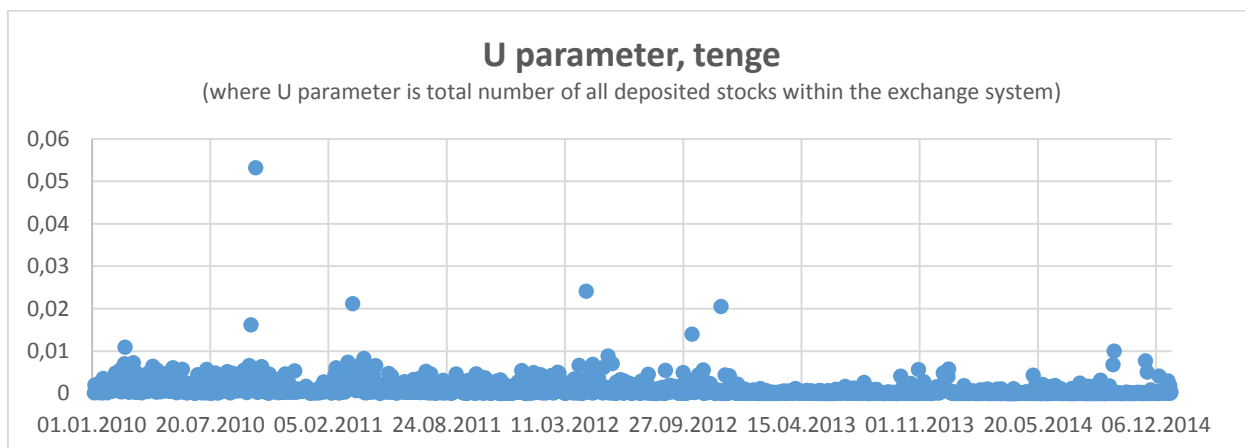
## Appendices 7-11. Results of the first approach calculations

Appendix 7

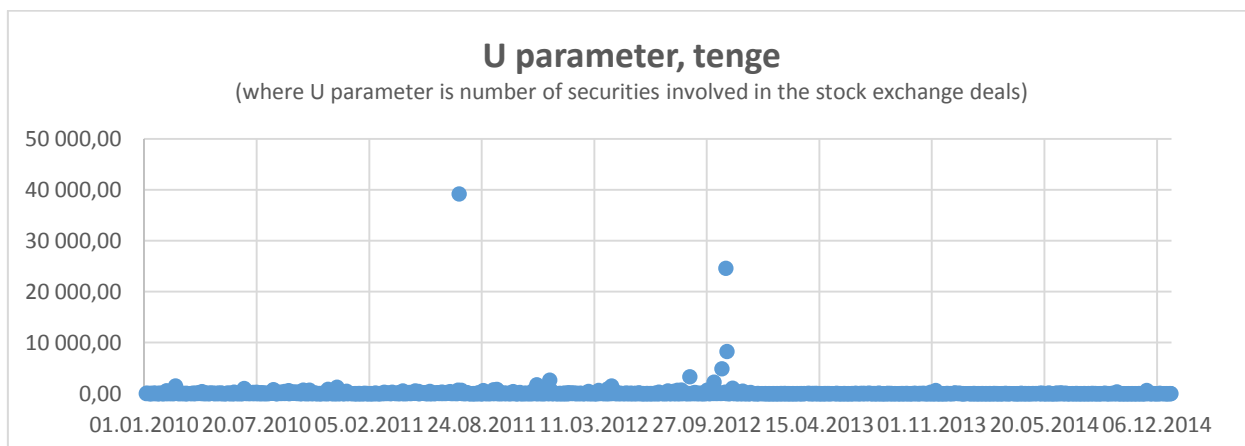
### U parameter calculations

|                           | Calculation,<br>where U parameter is number<br>of securities involved in the<br>stock exchange deals | Calculation,<br>where U parameter is total<br>number of all deposited<br>stocks within the exchange<br>system |
|---------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Arithmetic mean, tenge    | 0,0011                                                                                               | 124,90                                                                                                        |
| Standard deviation, tenge | 0,0024                                                                                               | 1 357,60                                                                                                      |

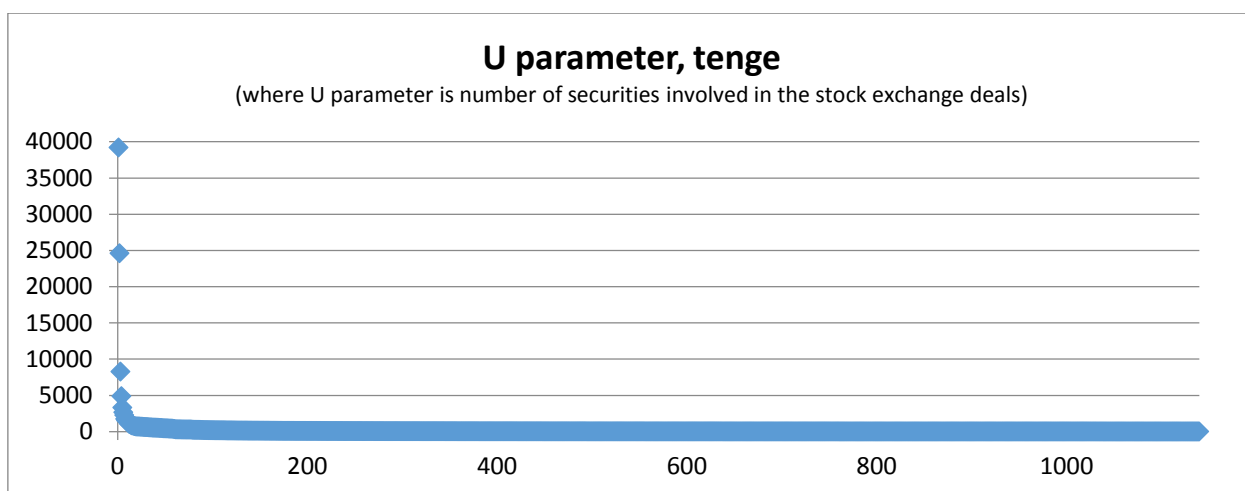
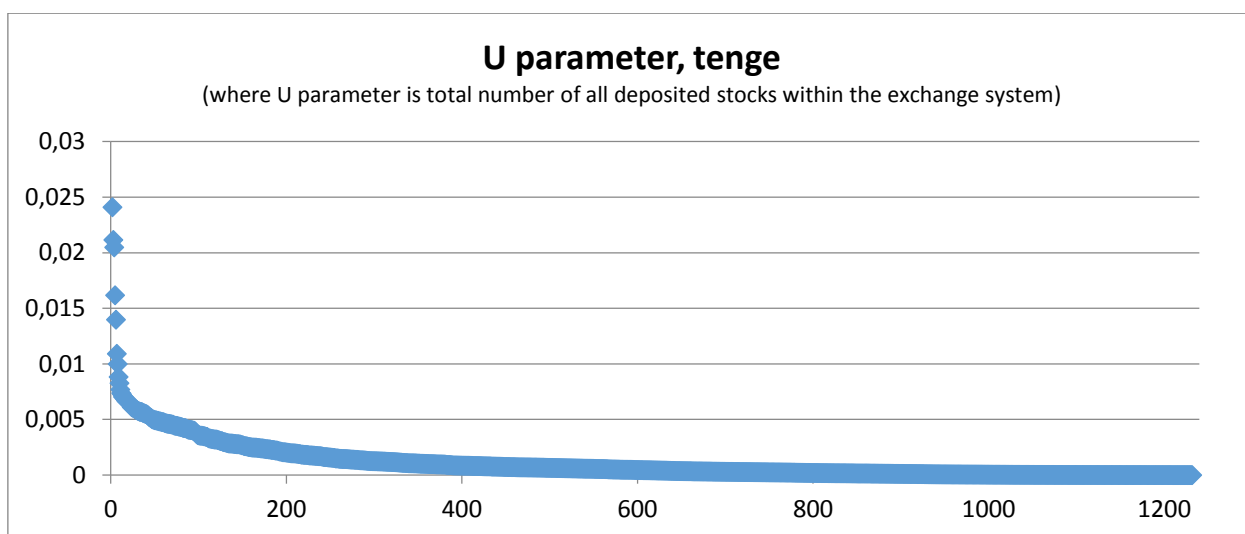
Appendix 8



Appendix 9





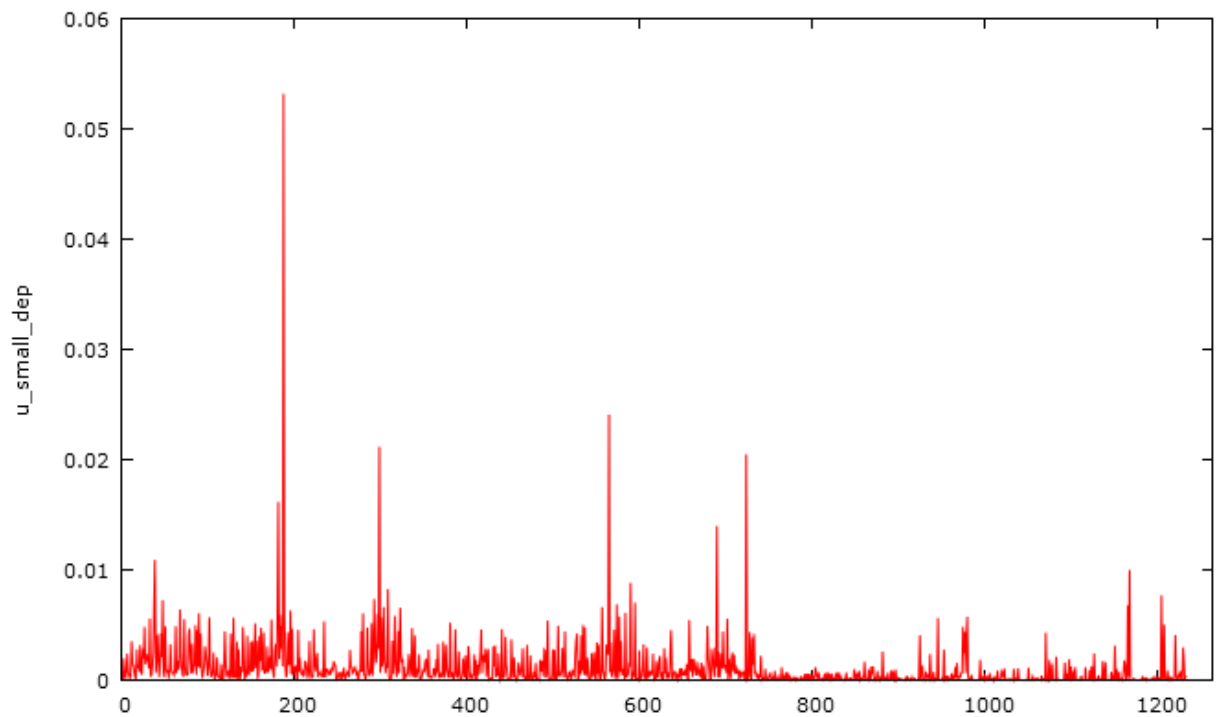


## Appendices 12-14. Results of the second approach calculations

Appendix 12

| Variable name in the theoretical model | Variable name in Gretl | Definition                                                                                         |
|----------------------------------------|------------------------|----------------------------------------------------------------------------------------------------|
| u                                      | u_small_dep            | Mean loss per a deal involving one stock (calculated using the amount of deposited stocks)         |
| u                                      | u_small_vol            | Mean loss per a deal involving one stock (calculated using the amount of stocks involved in deals) |
| I                                      | I                      | Volume of speculative investment (amount of money in the exchange's authorized bank)               |
| R                                      | R                      | Rate of one-day credits in the interbank market                                                    |
| U                                      | U_big_dep              | Total amount of deposited stocks within the exchange system                                        |
| U                                      | U_big_vol              | Total amount of stocks involved in the stock exchange deals                                        |

### 13. 1 Unit root test for u\_small\_dep



Augmented Dickey-Fuller test for u\_small\_dep  
 including 18 lags of  $(1-L)u\_small\_dep$  (max was 22)  
 sample size 1213  
 unit-root null hypothesis:  $a = 1$

test with constant

model:  $(1-L)y = b_0 + (a-1)y(-1) + \dots + e$

1st-order autocorrelation coeff. for e: -0.001

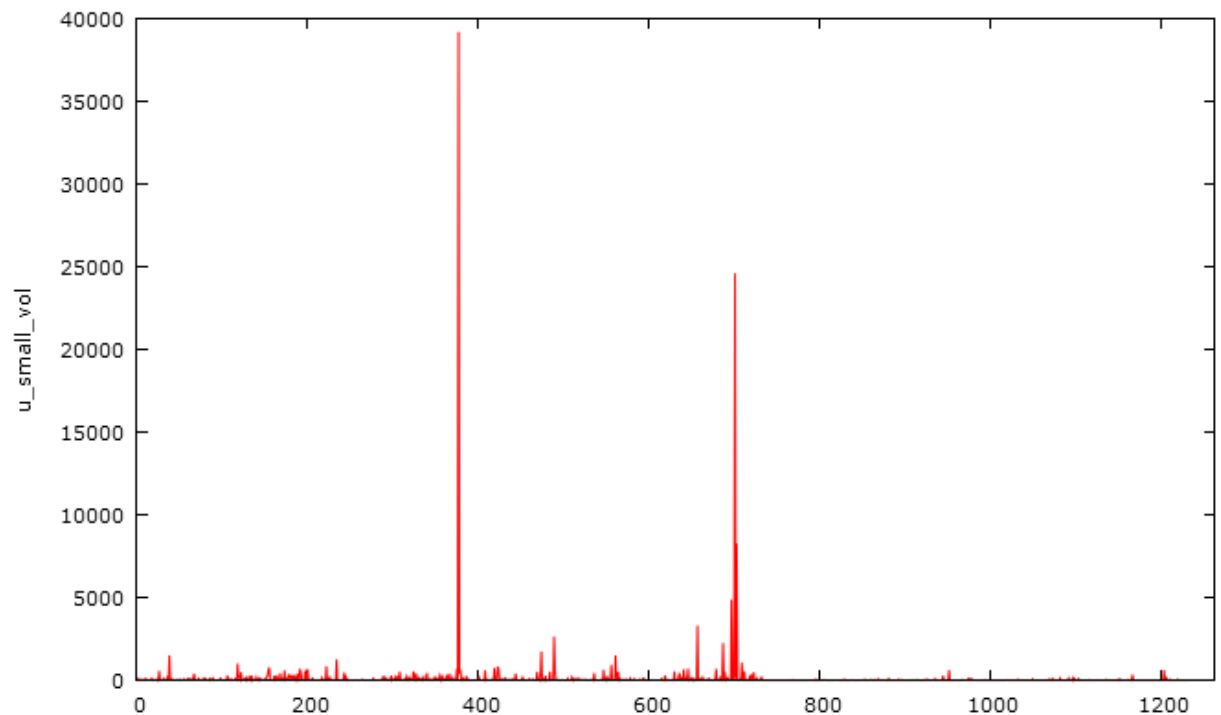
lagged differences:  $F(18, 1193) = 6.776$  [0.0000]

estimated value of  $(a - 1)$ : -0.363392

test statistic:  $\tau_c(1) = -5.24083$

asymptotic p-value 6.397e-006

### 13.2 Unit root test for u\_small\_vol



Augmented Dickey-Fuller test for u\_small\_vol  
including one lag of  $(1-L)u\_small\_vol$  (max was 22)  
sample size 1230  
unit-root null hypothesis:  $a = 1$

test with constant

model:  $(1-L)y = b_0 + (a-1)y(-1) + \dots + e$

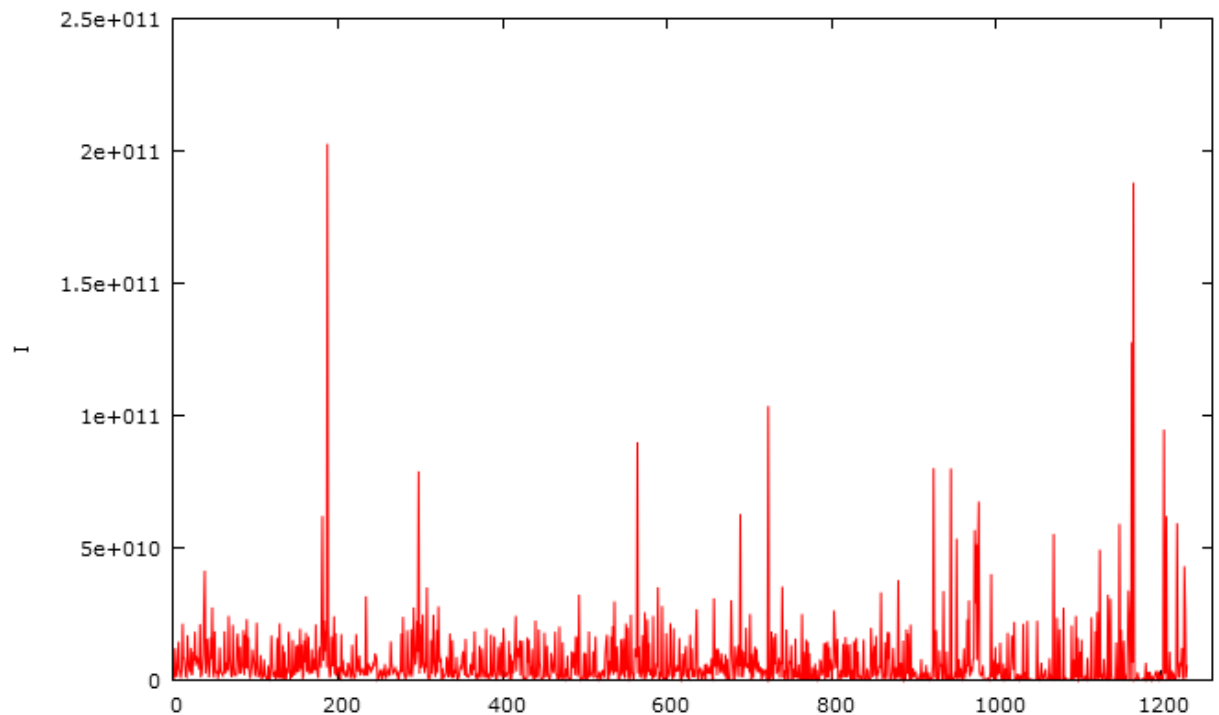
1st-order autocorrelation coeff. for e: 0.000

estimated value of  $(a - 1)$ : -0.886475

test statistic:  $\tau_c(1) = -22.1575$

asymptotic p-value 1.601e-050

### 13.3 Unit root test for I



Augmented Dickey-Fuller test for I  
including 17 lags of  $(1-L)I$  (max was 22)  
sample size 1214  
unit-root null hypothesis:  $\alpha = 1$

test with constant

model:  $(1-L)y = b_0 + (\alpha-1)y(-1) + \dots + e$

1st-order autocorrelation coeff. for e: -0.001

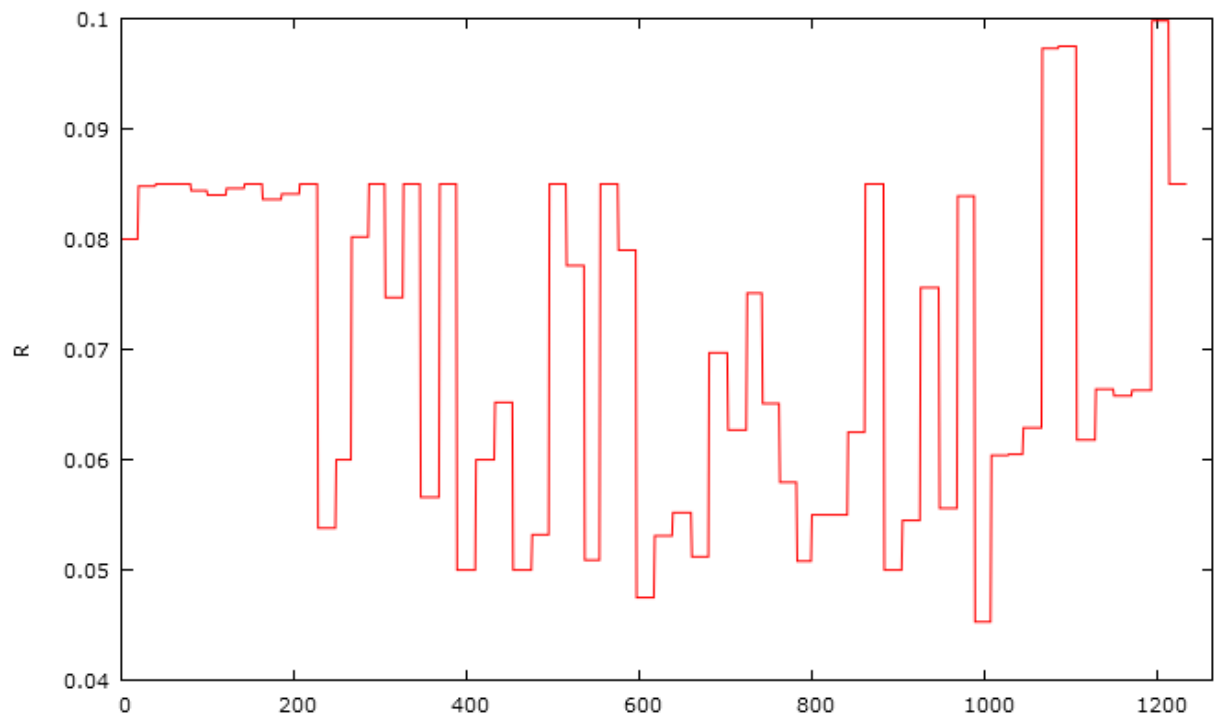
lagged differences:  $F(17, 1195) = 2.007$  [0.0088]

estimated value of  $(\alpha - 1)$ : -0.793051

test statistic:  $\tau_c(1) = -7.75998$

asymptotic p-value 2.491e-012

### 13.4 Unit root test for R



Augmented Dickey-Fuller test for R  
including 22 lags of  $(1-L)R$  (max was 22)  
sample size 1209  
unit-root null hypothesis:  $\alpha = 1$

test with constant

model:  $(1-L)y = b_0 + (\alpha-1)y(-1) + \dots + e$

1st-order autocorrelation coeff. for e: 0.001

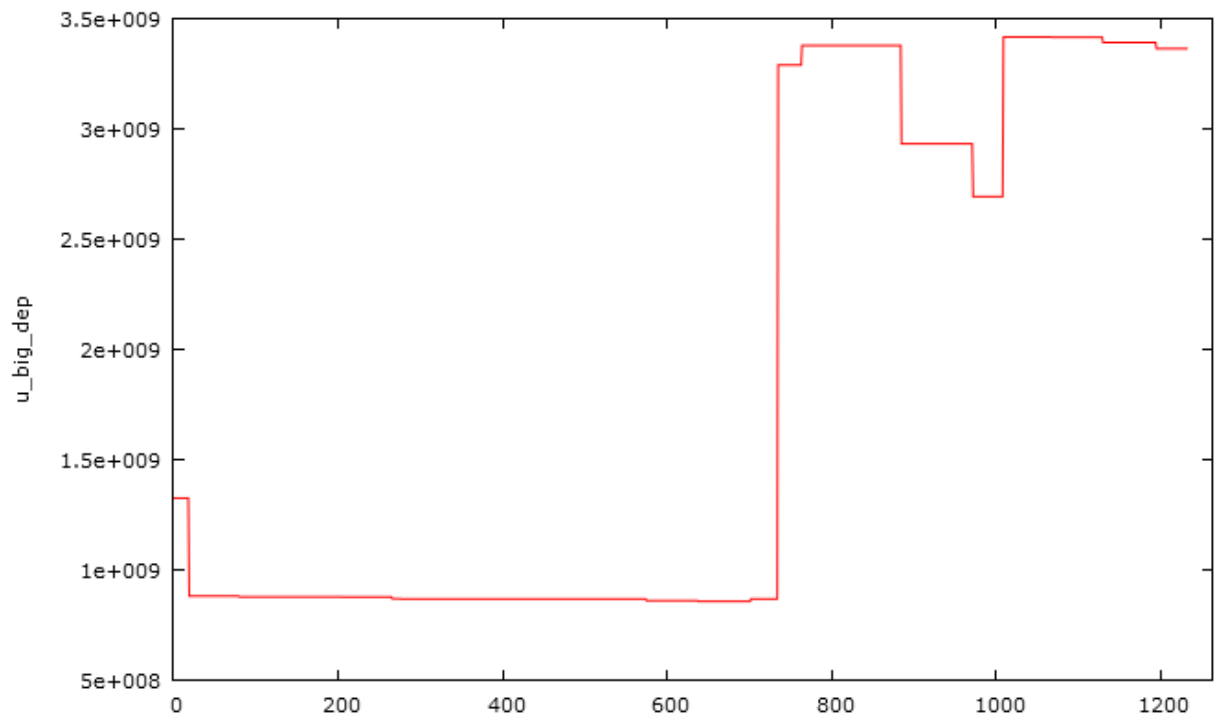
lagged differences:  $F(22, 1185) = 2.461$  [0.0002]

estimated value of  $(\alpha - 1)$ : -0.0349829

test statistic:  $\tau_c(1) = -3.67816$

asymptotic p-value 0.004453

### 13.5 Unit root test for u\_big\_dep



Augmented Dickey-Fuller test for u\_big\_dep

sample size 1231

unit-root null hypothesis:  $a = 1$

test with constant

model:  $(1-L)y = b_0 + (a-1)y(-1) + e$

1st-order autocorrelation coeff. for e: 0.000

estimated value of  $(a - 1)$ : -0.00140949

test statistic:  $\tau_c(1) = -0.778644$

p-value 0.8242

Augmented Dickey-Fuller test for d\_u\_big\_dep

sample size 1230

unit-root null hypothesis:  $a = 1$

test with constant

model:  $(1-L)y = b_0 + (a-1)y(-1) + e$

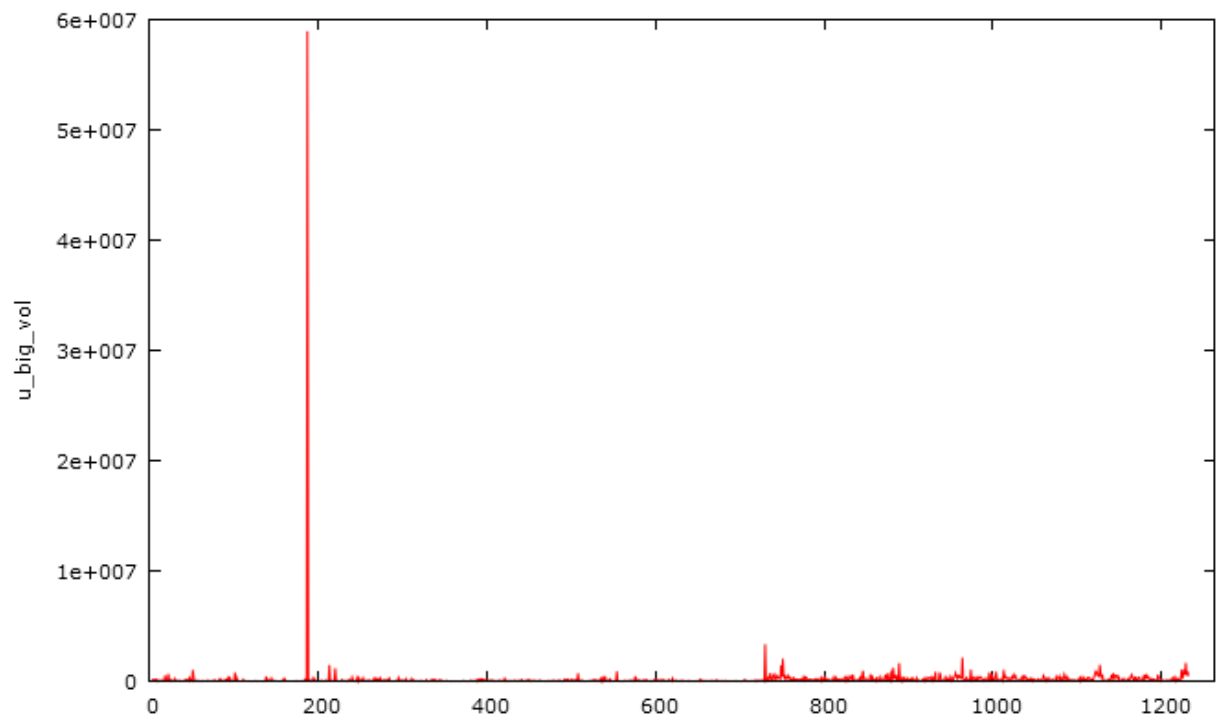
1st-order autocorrelation coeff. for e: -0.000

estimated value of  $(a - 1)$ : -1.00049

test statistic:  $\tau_c(1) = -35.0601$

p-value 9.696e-025

### 13.6 Unit root test for u\_big\_vol



Dickey-Fuller test for `u_big_vol`

sample size 1231

unit-root null hypothesis:  $a = 1$

test with constant

model:  $(1-L)y = b_0 + (a-1)y(-1) + e$

1st-order autocorrelation coeff. for `e`: -0.000

estimated value of  $(a - 1)$ : -1.00024

test statistic:  $\tau_c(1) = -35.0651$

p-value  $9.836 \times 10^{-25}$



**ADF test results summary:**

| Variable name in Gretl | ADF test result                                                             |
|------------------------|-----------------------------------------------------------------------------|
| u_small_dep            | Variable is stationary at the 1% level of significance                      |
| u_small_vol            | Variable is stationary at the 1% level of significance                      |
| I                      | Variable is stationary at the 1% level of significance                      |
| R                      | Variable is stationary at the 1% level of significance                      |
| U_big_dep              | Variable is stationary in first differences at the 1% level of significance |
| U_big_vol              | Variable is stationary at the 1% level of significance                      |

### 14.1 Calculation with amount of deposited stocks

#### Linear regression of $u\_small\_dep$ using $u\_big\_dep$ , I, R and constant

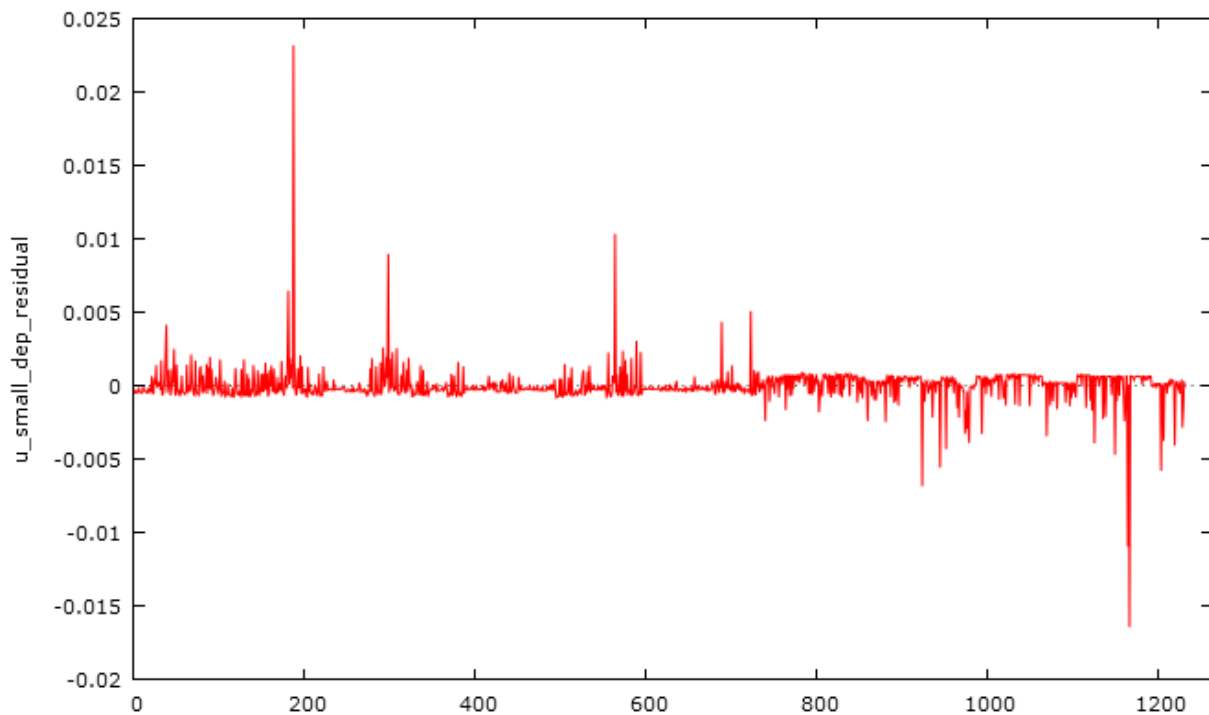
Model 1: OLS, using observations 1-1232

Dependent variable:  $u\_small\_dep$

|               | <i>Coefficient</i> | <i>Std. Error</i> | <i>t-ratio</i> | <i>p-value</i> |     |
|---------------|--------------------|-------------------|----------------|----------------|-----|
| const         | -5.39328e-05       | 0.000191808       | -0.2812        | 0.77862        |     |
| I             | 1.44129e-013       | 0                 | 52.9767        | <0.00001       | *** |
| R             | 0.0148303          | 0.00249189        | 5.9514         | <0.00001       | *** |
| $u\_big\_dep$ | -4.66606e-013      | 0                 | -15.0063       | <0.00001       | *** |

|                    |           |                    |           |
|--------------------|-----------|--------------------|-----------|
| Mean dependent var | 0.001137  | S.D. dependent var | 0.002413  |
| Sum squared resid  | 0.002000  | S.E. of regression | 0.001276  |
| R-squared          | 0.721051  | Adjusted R-squared | 0.720369  |
| F(3, 1228)         | 1058.078  | P-value(F)         | 0.000000  |
| Log-likelihood     | 6463.734  | Akaike criterion   | -12919.47 |
| Schwarz criterion  | -12899.00 | Hannan-Quinn       | -12911.77 |
| rho                | 0.014530  | Durbin-Watson      | 1.970819  |

#### ADF test results for residuals:



Augmented Dickey-Fuller test for  $u\_small\_dep\_residual$   
including 5 lags of  $(1-L)u\_small\_dep\_residual$  (max was 22)

sample size 1226

unit-root null hypothesis:  $a = 1$

test with constant

model:  $(1-L)y = b_0 + (a-1)y(-1) + \dots + e$

1st-order autocorrelation coeff. for e: -0.002

lagged differences:  $F(5, 1219) = 9.465$  [0.0000]

estimated value of  $(a - 1)$ : -0.671148

test statistic:  $\tau_c(1) = -11.0798$

asymptotic p-value 1.053e-022

## 14.2 Calculation with volume of trade

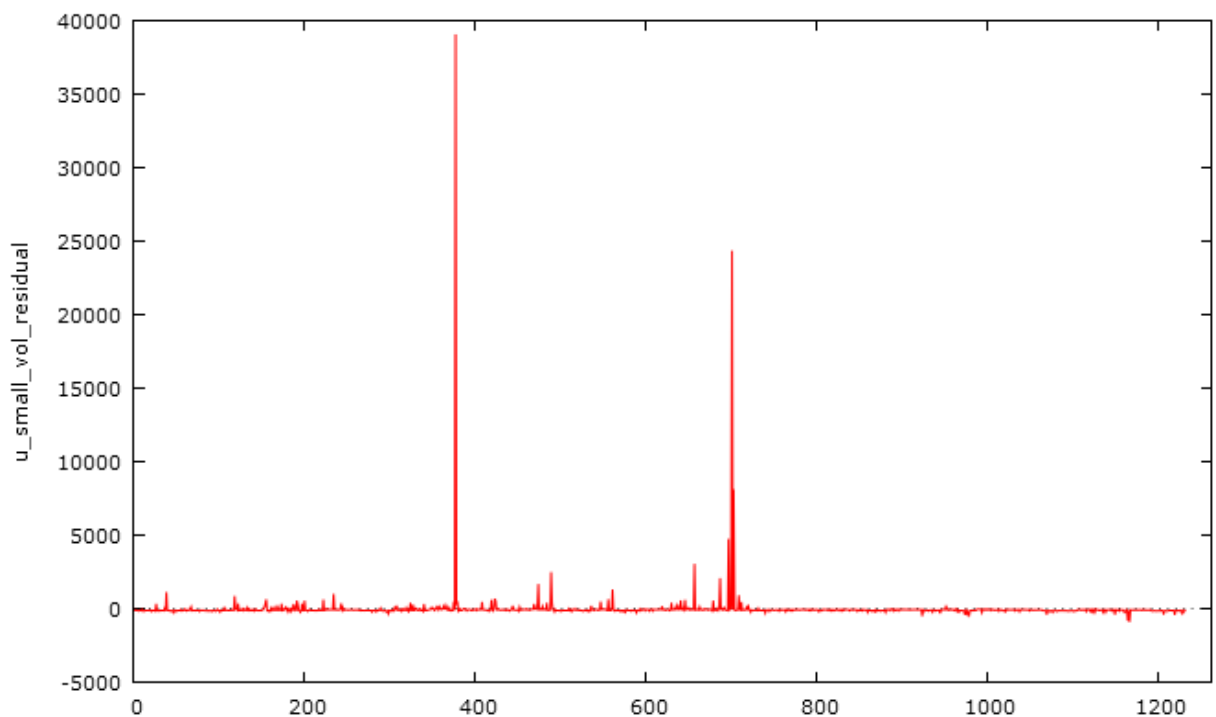
### Linear regression of u\_small\_vol using u\_big\_vol, I, R and constant

Model 2: OLS, using observations 1-1232

Dependent variable: u\_small\_vol

|                    | <i>Coefficient</i> | <i>Std. Error</i>  | <i>t-ratio</i> | <i>p-value</i> |   |
|--------------------|--------------------|--------------------|----------------|----------------|---|
| const              | -48.3371           | 188.06             | -0.2570        | 0.79720        |   |
| I                  | 6.18352e-09        | 3.17654e-09        | 1.9466         | 0.05181        | * |
| R                  | 1934.66            | 2636.76            | 0.7337         | 0.46325        |   |
| u_big_vol          | -2.8605e-05        | 2.51251e-05        | -1.1385        | 0.25513        |   |
| Mean dependent var | 124.9018           | S.D. dependent var | 1357.595       |                |   |
| Sum squared resid  | 2.26e+09           | S.E. of regression | 1356.684       |                |   |
| R-squared          | 0.003776           | Adjusted R-squared | 0.001342       |                |   |
| F(3, 1228)         | 1.551350           | P-value(F)         | 0.199535       |                |   |
| Log-likelihood     | -10632.30          | Akaike criterion   | 21272.59       |                |   |
| Schwarz criterion  | 21293.06           | Hannan-Quinn       | 21280.29       |                |   |
| rho                | 0.007715           | Durbin-Watson      | 1.984558       |                |   |

### ADF test results for residuals:



Augmented Dickey-Fuller test for u\_small\_vol\_residual

including one lag of  $(1-L)u\_small\_vol\_residual$  (max was 22)

sample size 1230

unit-root null hypothesis:  $a = 1$

test with constant

model:  $(1-L)y = b_0 + (a-1)y(-1) + \dots + e$

1st-order autocorrelation coeff. for e: 0.000

estimated value of  $(a - 1)$ : -0.886062

test statistic:  $\tau_c(1) = -22.1592$

asymptotic p-value 1.594e-050